# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A bag has $60$ balls, red or blue. If $P(\text{red}) = \frac{7}{10}$, the number of blue balls is: 1

**(a)** 20 **(b)** 19 **(c)** 17 **(d)** 18

**2.** A letter is chosen from the word EXAMINATION. The probability that it is neither A nor N is: 1

**(a)** $\frac{4}{11}$ **(b)** $\frac{7}{12}$ **(c)** $\frac{8}{11}$ **(d)** $\frac{7}{11}$

**3.** The probability that a seed germinates is $\frac{5}{8}$. Out of $4800$ seeds the expected number that will NOT germinate is: 1

**(a)** 1802 **(b)** 1801 **(c)** 1799 **(d)** 1800

**4.** If an experiment has $n$ equally likely outcomes, the probabilities of all outcomes add up to: 1

**(a)** $1$ **(b)** $0$ **(c)** $\frac{1}{n}$ **(d)** $n$

**5.** Two dice are rolled together. The probability that the product of the numbers is 9 is: 1

**(a)** $\frac{1}{18}$ **(b)** $\frac{1}{36}$ **(c)** $\frac{1}{37}$ **(d)** $\frac{35}{36}$

**6.** Two dice are rolled together. The probability that the sum is divisible by 4 is: 1

**(a)** $\frac{1}{5}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

**7.** Three coins are tossed. The probability of getting more heads than tails is: 1

**(a)** $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

**8.** A bag has $9$ red balls and some blue balls. If the probability of drawing a blue ball is $\frac{4}{7}$, the number of blue balls is: 1

**(a)** 11 **(b)** 12 **(c)** 14 **(d)** 13

**9.** A bag has $84$ balls, red or blue. If $P(\text{red}) = \frac{3}{7}$, the number of blue balls is: 1

**(a)** 48 **(b)** 49 **(c)** 50 **(d)** 47

**10.** A letter is chosen from the word PROBABILITY. The probability that it is a consonant that is not B is: 1

**(a)** $\frac{5}{12}$ **(b)** $\frac{5}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{6}{11}$

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Find the probability that the larger of the two numbers on rolled dice is exactly 5. 2

**12.** Two dice are rolled together. Find the probability that the product of the numbers is 5. 2

**13.** A bag has 18 red balls and some blue balls. If $P(\text{blue}) = \frac{4}{7}$, find the number of blue balls. 2

**14.** Find the probability that the larger of the two numbers on rolled dice is exactly 6. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A bag contains red, blue and green balls. $P(\text{red}) = \frac{3}{12}$ and $P(\text{blue}) = \frac{4}{12}$ and there are $60$ balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

**16.** In 80 throws of a die, the number of times a prime number appeared was $41$ and a number greater than 4 appeared $25$ times (these include $11$ throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

**17.** A letter is chosen at random from the word CHANDIGARH. Find the probability that it is 3
(i) not H,
(ii) a vowel or H.

**OR**

A bag contains red, blue and green balls. $P(\text{red}) = \frac{2}{9}$ and $P(\text{blue}) = \frac{5}{9}$ and there are $54$ balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

**18.** In 250 throws of a die, the number of times a prime number appeared was $130$ and a number greater than 4 appeared $84$ times (these include $39$ throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

**OR**

A letter is chosen at random from the word RAJASTHAN. Find the probability that it is 3
(i) not A,
(ii) a vowel or A.

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Two dice are rolled together. Find the probability that 5
(i) the sum is 5,
(ii) the sum is more than 5,
(iii) the sum is at most 5. Verify that
(ii) and
(iii) add to $1$.

**OR**

A bag has 5 red, 9 blue and 16 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{3}$. How many green balls were removed? 5

**20.** Two dice are rolled together. Find the probability that 5
(i) the sum is 10,
(ii) the sum is more than 10,
(iii) the sum is at most 10. Verify that
(ii) and
(iii) add to $1$.

**OR**

A bag has 8 red, 8 blue and 14 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{5}$. How many green balls were removed? 5

## END OF QUESTION PAPER — Total: 40 Marks

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